

MUHAMMAD RAFAY KHAN

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PROFILE

Financial Technology undergraduate (BS FinTech, Mohammad Ali Jinnah University) with hands-on project experience in machine learning, statistical analysis, Islamic finance structuring, and database design. Particular interest in insurtech, claims analytics, and data-driven financial decision-making.

EDUCATION

Mohammad Ali Jinnah University (MAJU), Karachi

BS Financial Technology — 6th Semester (Expected 2028)

Jinnah Government College

Higher Secondary School Certificate (Pre-Engineering) — 2022

Little Folks Paradise School

Secondary School Certificate (Science) — 2020

ACADEMIC PROJECTS

MCB Bank Stock Prediction — Random Forest ML Model — *AI in Financial Decision, Group Project*

- Built and iteratively refined a Random Forest model predicting next-day MCB Bank (PSX) stock returns, engineering 89 features across price momentum, technical indicators, macroeconomic variables, and sector data
- Achieved 56.6% walk-forward directional accuracy (13 of 15 quarters above baseline) and a backtested Sharpe ratio of 2.03; validated results using SHAP, permutation, and MDI feature-importance methods

Tools: Python, scikit-learn, time-series cross-validation

Diminishing Musharakah Financing Model — *Islamic Finance Case Study, Group Project*

- Modeled a PKR 1 billion Shariah-compliant commercial property financing structure for a PSX-listed company, building a full 7-year rental-sharing and equity buyback schedule
- Conducted a 6-category risk assessment with Shariah-compliant mitigations and benchmarked the structure against conventional loan financing

Tools: Financial modeling, Excel

Predictive Modeling on Insurance & Customer Data — *Data Science & Machine Learning, Course Project*

- Built a linear regression model predicting insurance charges from policyholder data, supported by exploratory data analysis covering distribution, outlier, and correlation analysis
- Applied K-Nearest Neighbors with feature scaling to a customer dataset for similarity and pattern analysis

Tools: Python, pandas, scikit-learn, seaborn, matplotlib

Financial Literacy & Saving Behavior — *Statistical Research Study, Group Project*

- Co-designed and fielded a 15-item survey (n = 86) measuring financial literacy and saving behavior among university students
- Ran reliability (Cronbach's α), normality, correlation, and regression tests in SPSS; found financial literacy explains 46% of variance in saving behavior ($R^2 = 0.462$, $p < 0.001$)

Tools: SPSS, survey design, hypothesis testing

Relational Database Design — School Management System — *Database Systems, Course Project*

- Designed a 22-table relational database schema covering attendance, library, fee, examination, transportation, and hostel-management modules
- Implemented primary-key-based table structures and populated the schema with sample records for query testing

Tools: SQL (T-SQL / SQL Server)

SKILLS

Technical: Python (pandas, scikit-learn, seaborn, matplotlib), SQL, SPSS, MS Excel

Financial: Financial & cost accounting fundamentals, statistical & regression analysis, Islamic finance structuring

Core: Analytical thinking, research design, presentation & communication, teamwork

INTERESTS

Financial markets, digital banking systems, insurtech & claims analytics